ARAVINDHSAMY P

Biomedical Engineering Student | Embedded IoMT Developer

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PROFESSIONAL SUMMARY

Detail-oriented Biomedical Engineering undergraduate (CGPA 7.80, graduating 2027) with hands-on experience in embedded systems, IoMT and medical image processing. Developed Arduino-based healthcare devices with GSM/SMS connectivity and gained practical exposure through internships in hospital patient monitoring, biomedical equipment, embedded IoT with biomedical sensors, and MATLAB-based image analysis. Seeking a Biomedical or Embedded Systems Engineering role to contribute to reliable, low-cost medical technology.

TECHNICAL SKILLS

Biomedical Instrumentation: Patient monitoring systems, biomedical equipment handling, biomedical sensors
Biosignals & Sensing: ECG, SpO_2 (pulse oximetry), temperature sensing, signal acquisition
Medical Image Processing: MATLAB-based image analysis and processing
Embedded & IoMT: Arduino Uno, ESP32, GSM (SMS alerts), IoT connectivity
Tools: MATLAB, Arduino IDE, Git, GitHub, VS Code

PROJECTS

Smart Medicine Reminder Kit (Arduino Uno / GSM / SMS)
- Designed a medicine reminder kit as a third-year project using Arduino Uno to help patients follow dosage schedules.
- Integrated a GSM module to send SMS reminders and alerts to the patient and caregiver.
- Documented the system architecture with a block diagram and technical report.

Smart Medical Box (Arduino Uno Dispenser)
- Built an Arduino Uno-based automated medicine dispenser for scheduled, controlled dosing.
- Prepared full project documentation: component tables, table of contents and IEEE-format references.

CPNAS – Technical Report
- Authored a detailed 60-slide technical report covering design, methodology and results.

INTERNSHIP EXPERIENCE

Intern — Kauvery Hospital
- Gained practical exposure to patient monitoring systems and biomedical equipment in a clinical environment.

Intern — Neuro Company Private Limited
- Worked on embedded IoT systems and biomedical sensors.

Intern — Medical Image Processing, Salem
- Worked on MATLAB-based medical image processing.

EDUCATION

Bachelor of Engineering in Biomedical Engineering — Expected 2027
Annapoorna Engineering College (Anna University) | CGPA: 7.80

SOFT SKILLS

- Teamwork in Clinical / Healthcare Settings
- Analytical Problem Solving
- Attention to Detail & Patient-Safety Awareness
- Time Management
- Clear Technical Communication
- Fast Learning & Adaptability

CERTIFICATIONS

- Participation Certificate — IoMT Sensor & Microcontroller Learning, Neuro Technology
- Internship Certificate — Embedded Systems with IoMT, Chennai (1 Month)
- Participation Certificate — MATLAB Medical Image Processing

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.